MODULE 05 · SECTION 1 - FUNDAMENTALS OF GEN AI

Inside a foundation model, without the maths

Tokens, context windows, training, tuning, diffusion and thinking, explained for leaders.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 5 of 33

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A language model does one thing, over and over: **predict the next token**.

From prompt to answer

01
Split
The prompt is broken into tokens.

02
Read
The model takes in the whole context at once.

03
Predict
It chooses a likely next token.

04
Repeat
That token is added, and it predicts again.

05
Stop
It ends at a natural finish or a length limit.

THE TAKEAWAY

Every answer is built token by token. That is why length, cost and speed all depend on tokens.

Tokens: the unit everything is measured in

- **A token** is a piece of text: a character, part of a word, or a whole word.
- **Rule of thumb for Gemini:** about 4 characters per token; 100 tokens is 60 to 80 words.
- **Images, audio and video** are turned into tokens too.
- **Billing** counts tokens in and tokens out.
- **Long prompts and long answers** both cost more and take longer.

> ON THE EXAM
>
> When a question links cost or speed to the size of the input or output, it is talking about tokens.

The context window: how much it can hold at once

- **The context window** is the maximum number of tokens a model can consider at once.
- **Google's analogy:** short-term memory. Think of the space on a desk.
- **Many Gemini models** take 1 million tokens or more: about eight novels, or 50,000 lines of code.
- **Use it for** long contracts, many documents, long videos, whole codebases.
- **Limits:** finding many separate facts gets less accurate, and big inputs are slower.

> ON THE EXAM
>
> Long input that must be considered all at once? The deciding feature is a large context window.

Context window or output length?

CONTEXT WINDOW

How much it can read

- Prompt, documents, chat history
- Measured in input tokens
- Matters for long inputs
- A property of the model

OUTPUT LENGTH

How much it may write

- The answer it generates
- Capped by a max output tokens setting
- Matters for long or short replies
- A setting you choose per request

THE TIEBREAKER

Reading a huge document is a context-window question. Getting a shorter answer is an output-length question.

Parameters: what the model learned

- **Parameters** are the numbers inside the model, set during training.
- **Large models** have billions of them. That is the large in large language model.
- **More capacity** usually means stronger reasoning and broader knowledge.
- **The trade-off:** bigger models are slower and cost more per token.
- **Hence tiers:** Gemini Pro for hard tasks, Flash and Flash-Lite for speed and scale.

> THE TIEBREAKER
>
> The biggest model is not automatically the best choice. Fit the size to the task. Module 9 goes further.

The life of a foundation model

01

Pre-training

Vast unlabelled data; enormous compute. Done by the model maker.

02

Shaping

Tuned on examples and human feedback to follow instructions.

03

Adapting

You prompt it, ground it in your data, or tune it.

04

Inference

It answers your requests, one token at a time.

ON THE EXAM

Organisations almost never pre-train. They pick a model, then adapt it, at stage three.

Four ways to adapt a model, lightest first

APPROACH	WHAT CHANGES, AND THE EFFORT
Prompt engineering	Only your instructions. No training. Minutes to try.
Grounding and RAG	The model is handed your data with each request. No training.
Prompt tuning	A small learned prefix is trained; the model itself is frozen.
Fine-tuning	The model's own parameters are adjusted on labelled examples.

THE TIEBREAKER

Start at the top and move down only when you must. Each step down costs more data, skill and time.

Prompt engineering or prompt tuning?

PROMPT ENGINEERING

You write better instructions

- No training at all
- Words, examples, roles, format
- Anyone can do it, today
- Covered in module 20

PROMPT TUNING

A small component is trained

- Needs labelled examples
- Learns a prefix; model frozen
- Cheaper than full fine-tuning
- A form of parameter-efficient tuning

THE TIEBREAKER

Engineering is writing. Tuning is training. The names are close; the effort is not.

Training or inference: who pays, and when?

TRAINING

Building the model

- Happens once, or occasionally
- Huge compute: thousands of TPUs
- Paid by the model maker
- You pay only if you tune

INFERENCE

Using the model

- Happens on every request
- Priced per token, in and out
- Paid by you, as you use it
- Grows with every user

THE TAKEAWAY

For most organisations, the running cost of generative AI is inference, driven by tokens and volume.

Why a fluent model can still be wrong

- **It predicts what sounds right,** not what has been checked.
- **No built-in doubt:** it rarely says I don't know unless asked to.
- **Frozen knowledge:** it knows nothing after its training cutoff.
- **Gaps get filled** with plausible detail: an invented date, a made-up policy.
- **The fix** is to give it facts at request time: grounding, in module 21.

> ON THE EXAM
>
> Confident, fluent and wrong is called a hallucination. Next-token prediction is why it happens.

Diffusion: how images and video are generated

- **Start from random noise,** like a screen of static.
- **Remove the noise step by step,** guided by the text prompt.
- **A clear image or video** emerges that matches the description.
- **Used for** images and video, such as Imagen and Veo.
- **Not** for predicting labels or storing data. It generates media.

> ON THE EXAM
>
> Diffusion turns noise into a structured image, guided by a prompt. That is the one-line definition to recognise.

Thinking models: reasoning before answering

- **Thinking models** reason internally before they reply.
- **Better at** multi-step problems: planning, analysis, maths, code.
- **Recent Gemini models** think by default and adjust the effort themselves.
- **A thinking level** setting turns the effort up or down.
- **Thinking costs** extra tokens and time, and those tokens are billed.

> THE TIEBREAKER
>
> Hard, multi-step task: more thinking. Simple, high-volume task: less thinking, faster and cheaper.

Wrenfield's Dutch catalogue, in tokens

STEP	THE ARITHMETIC
Product descriptions to translate	48,000
Average description	About 150 words, roughly 200 tokens
Tokens per item, in and out	About 400
Total for the whole job	About 19 million tokens
Training needed	None. A clear prompt does it.

THE TAKEAWAY

Thinking in tokens turns a vague AI project into a sized, costed job a finance team can check.

A law firm wants to ask questions across a 900-page bundle of contracts, schedules and amendments.

Answers often depend on clauses hundreds of pages apart, so the whole bundle must be considered in one request.

Which model characteristic matters most?

A. A higher temperature setting

B. A larger maximum output length

C. A large context window

D. A diffusion-based architecture

◆ PAUSE AND DECIDE

Which model characteristic matters most?

A A higher temperature setting

Changes creativity, not how much the model can read.

B A larger maximum output length

Controls the answer's length, not the input's.

C A large context window

Lets the whole 900 pages be considered at once.

D A diffusion-based architecture

Diffusion generates images and video.

THE SIGNAL

The whole bundle in one request: input size, so the context window.

> Wrenfield's marketing team is moving to a warmer, more playful brand voice from tomorrow. Product copy is drafted with Gemini.
>
> The team has a two-page tone guide and a handful of approved examples. It has no data scientists.

What is the best first step?

A Fine-tune Gemini on 10,000 rewritten descriptions

B Put the tone guide and examples into the prompt

C Prompt-tune a prefix on labelled examples

D Pre-train a new model on the brand's archive

◆ PAUSE AND DECIDE

What is the best first step?

A Fine-tune Gemini on 10,000 rewritten descriptions

Those examples do not exist, and tuning takes skills and time.

B Put the tone guide and examples into the prompt

Prompt engineering: no training, ready tomorrow.

C Prompt-tune a prefix on labelled examples

Still training, with data the team does not have.

D Pre-train a new model on the brand's archive

Months of work and enormous cost, for a tone change.

THE SIGNAL

From tomorrow, a handful of examples, no data scientists: start at the top of the ladder.

> An advertising agency is pitching AI-generated campaign images to a cautious client.
>
> The client asks how the image model actually produces a picture from a sentence.

Which description of a diffusion model is accurate?

A. It predicts the next word, one at a time, from left to right

B. It starts from noise and refines it, step by step, towards the prompt

C. It finds the closest match in a library of stock photos

D. It sorts existing images into labelled categories

◆ PAUSE AND DECIDE

Which description of a diffusion model is accurate?

A It predicts the next word, one at a time, from left to right

That describes a language model.

B It starts from noise and refines it, step by step, towards the prompt

Noise in, image out, guided by the text.

C It finds the closest match in a library of stock photos

That is search. A diffusion model creates a new image.

D It sorts existing images into labelled categories

That is classification, a predictive task.

THE SIGNAL

Each wrong option is a true description of a different technique. Name the neighbour.

> A bank's app sorts two million short customer messages a day into three categories: payment, card or other. It uses a Gemini model with a high thinking level.
>
> Accuracy is fine, but cost and response times are far above target.

What is the most sensible change?

A Raise the thinking level further to protect accuracy

B Move to the largest, most capable model tier

C Lower the thinking level, or use a faster, lighter model

D Set a very low output limit so thinking is cut short

◆ PAUSE AND DECIDE

What is the most sensible change?

A Raise the thinking level further to protect accuracy

More cost and delay, for accuracy that is already fine.

B Move to the largest, most capable model tier

Slower and dearer per token. Wrong direction.

C Lower the thinking level, or use a faster, lighter model

A simple, high-volume task needs little reasoning.

D Set a very low output limit so thinking is cut short

Can leave answers incomplete, and cut-off thinking is still billed.

THE SIGNAL

Simple task, huge volume, accuracy already fine: spend less effort per request.

Module 5 in five lines

- Language models predict the next token; cost and speed follow tokens in and out.
- Context window is how much it reads; output length is how much it writes.
- Adapt lightest first: prompt, ground, prompt-tune, fine-tune. Organisations rarely pre-train.
- Diffusion turns noise into images and video, guided by a prompt.
- Fluent is not the same as checked: hallucinations come from next-token prediction.